

Data Science and Analytics

Chapter 4: Data Classification

1. Objectives

2. Data Classification Problems

3. Data Classification Models

4. Building a Classification Model

5. Example 1: Iris Dataset

6. Example 2: Titanic Dataset

7. Example 3: Breast Cancer Wisconsin Diagnostic Dataset

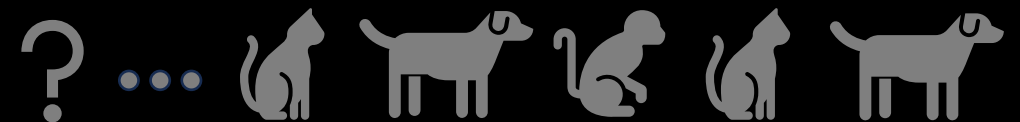
8. Summary

Objectives

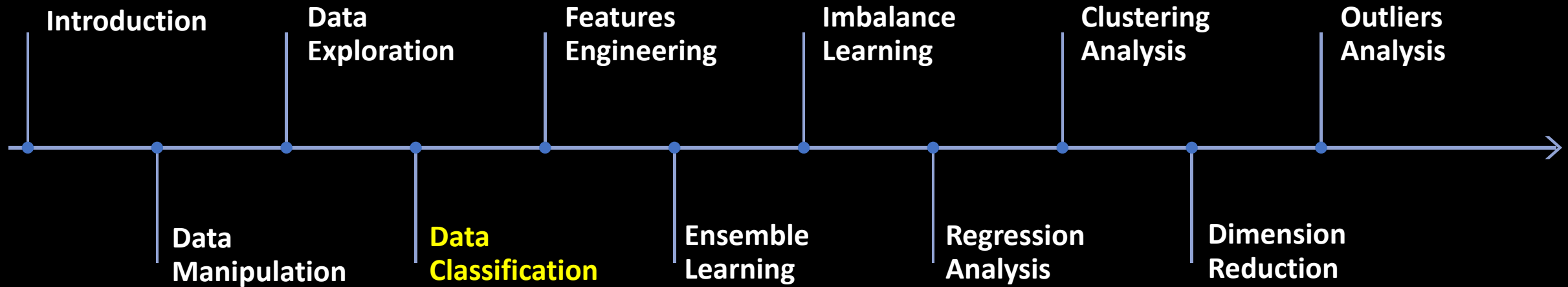
You should be able to answer the following questions:

- What is a data classification problem?
- Why can machine learning solve classification problems?
- What are techniques and tools for classification?
- How to create classification models efficiently?

“Classification is the art of turning raw data into
meaningful decisions”

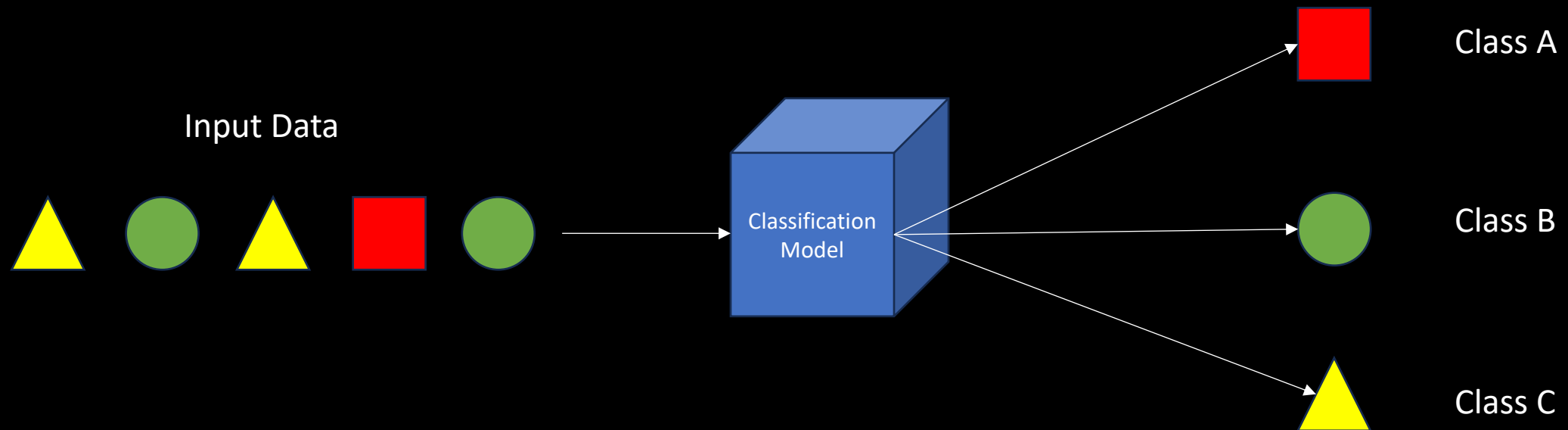


Current milestone



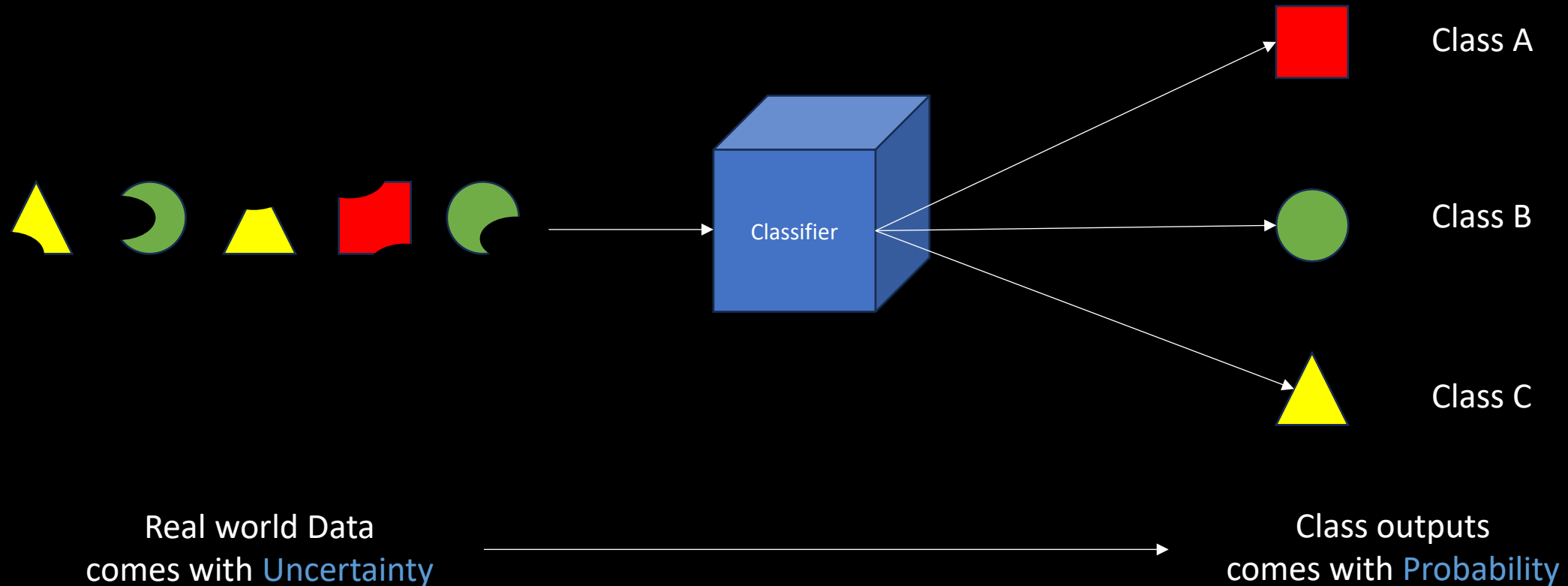
What is the classification problem?

- A classification problem is a type of problem where the goal is to predict the categorical labels of new instances based on past observations. Essentially, the model learns from the data which features are associated with which categories or classes.



What is uncertainty in classification?

- Uncertainty in classification refers to the degree of doubt or lack of confidence in assigning a particular class label to an instance. It's an important concept in machine learning and statistical classification, where models often predict not just the class label, but also a probability or confidence score associated with that label.



Data Classification Problem (Cont.)

- **Email Classification:** Used to separate spam emails from non-spam emails.
 - Class 0 - non-spam
 - Class 1 - Spam
- **Image Classification:** Used to categorize objects in images, such as distinguishing different animals in photographs.
 - Class 0 - Cat
 - Class 1 - Dog
 - Class 2 - Others
- **Disease Classification:** Used to diagnose diseases from medical data, such as identifying breast cancer from mammography images.
 - Class 0 - Negative
 - Class 1 - Positive
- **Sentiment Analysis:** Used to categorize the sentiment of opinions or reviews, such as differentiating positive and negative reviews of products.
 - Class 0 – Negative
 - Class 1 – Positive
- **Customer Churn Analysis:** Predict whether a customer will stay or leave the service based on usage behavior such as purchase frequency, membership duration, and promotion usage.
 - Class 0 – Stay
 - Class 1 – Churn

Data Classification Problem (Cont.)

- **Fraud Detection:** Predict whether a financial transaction is legitimate or fraudulent based on features such as transaction amount, time, location, device used, and customer history.
 - Class 0 – Legitimate Transaction
 - Class 1 – Fraudulent Transaction
- **Credit Scoring:** Predict whether a loan applicant is a good risk or a bad risk based on financial and demographic features such as income, employment status, credit history, debt ratio, and loan amount.
 - Class 0 – Good Credit (likely to repay)
 - Class 1 – Bad Credit (likely to default)
- **Network Intrusion Detection:** Predict whether incoming network traffic is normal or malicious based on features such as packet size, protocol type, connection duration, and number of failed login attempts.
 - Class 0 – Normal Traffic
 - Class 1 – Intrusion / Attack

Key Points:

Categorical outcomes: The prediction is a label, not a continuous numeric value.

Defined categories or classes: The output belongs to discrete groups

Labeled training data: The dataset includes examples with known class labels.

Feature-based separation: Input features

Data Classification Problem (Cont.)

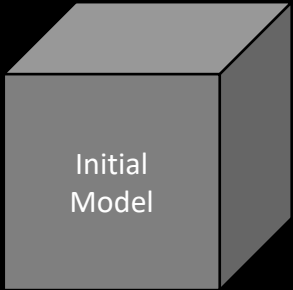
Training Dataset

Feature1	Feature2	Feature3	...	FeatureN	Label
12.87	0.5	M	...	True	Class 1
15.99	0.3	F	...	True	Class 2
45.68	0.8	M	...	False	Class 1
...	...	...	...	...	...
12.88	0.7	F	...	True	Class 2

Real-world Data

Feature1	Feature2	Feature3	...	FeatureN	Label
16.57	0.6	M	...	True	
31.48	0.4	M	...	False	
45.68	0.9	F	...	Ture	
...	...	...	...	...	

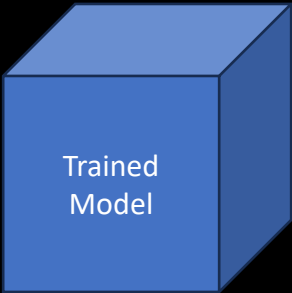
Train Model



Support Tools



Deploy Model



Make Prediction

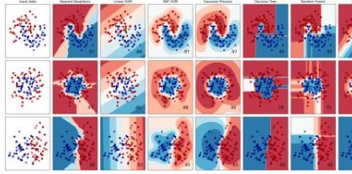
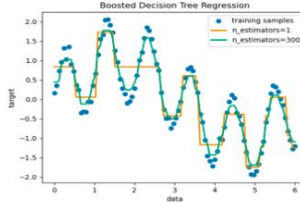
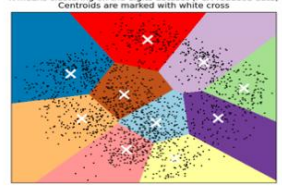
Label
Class 1
Class 1
Class 2
...

Data Classification Model

- Machine Learning Models for classification problem.

- Linear Models
- Linear and Quadratic Discriminant Analysis
- Kernel ridge regression
- Support Vector Machines**
- Stochastic Gradient Descent
- Nearest Neighbors**
- Gaussian Processes
- Cross decomposition
- Naive Bayes**
- Decision Trees**
- Ensembles: Gradient boosting, random forests, bagging, voting, stacking
- Multiclass and multioutput algorithms
- Feature selection
- Semi-supervised learning
- Isotonic regression
- Probability calibration
- Neural network models

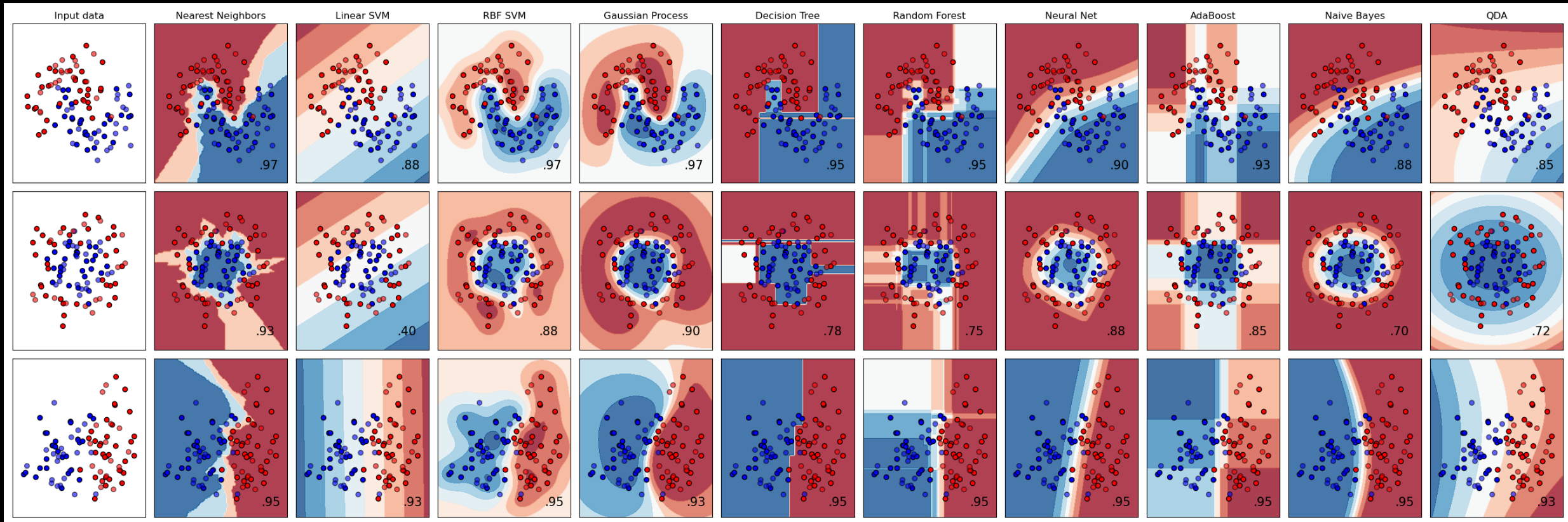
The screenshot shows the scikit-learn website with the following content:

- scikit-learn**
Machine Learning in Python
- Navigation links: Install, User Guide, API, Examples, Community, More
- Buttons: Getting Started, Release Highlights for 1.4, GitHub
- Key features:
 - Simple and efficient tools for predictive data analysis
 - Accessible to everybody, and reusable in various contexts
 - Built on NumPy, SciPy, and matplotlib
 - Open source, commercially usable - BSD license
- Classification**
Identifying which category an object belongs to.
Applications: Spam detection, image recognition.
Algorithms: Gradient boosting, nearest neighbors, random forest, logistic regression, and more...

Examples
- Regression**
Predicting a continuous-valued attribute associated with an object.
Applications: Drug response, Stock prices.
Algorithms: Gradient boosting, nearest neighbors, random forest, ridge, and more...

Examples
- Clustering**
Automatic grouping of similar objects into sets.
Applications: Customer segmentation, Grouping experiment outcomes
Algorithms: k-Means, HDBSCAN, hierarchical clustering, and more...

Examples
- Dimensionality reduction**
Reducing the number of random variables to
- Model selection**
Comparing, validating and choosing parameters and
- Preprocessing**
Feature extraction and normalization.

<https://scikit-learn.org/stable/>

Data Classification Model (Cont.)

- How does each classification models provide output?



https://scikit-learn.org/stable/auto_examples/classification/plot_classifier_comparison.html

Decision Trees

A non-parametric supervised learning method used for classification and regression. The goal is to create a model that predicts the value of a target variable by learning simple decision rules inferred from the data features.

Components:

- **Root Node** - The starting point; represents the entire dataset before any split.
- **Decision Nodes** - Internal nodes where the dataset is split based on a feature condition.
- **Branches** - Paths that connect nodes, representing outcomes of a decision rule.
- **Leaf Nodes** - Final nodes that represent the predicted class or value.
- **Splitting Criteria** - Rules used to decide how to split (e.g., Gini Index, Entropy, Information Gain).

Decision tree trained on all the iris features



<https://scikit-learn.org/stable/modules/tree.html#>

Create Decision Trees

1. **Start at the Root Node:** Begin with the entire dataset, the root node represents all examples.
2. **Select the Best Feature:** Choose the feature that best separates the data, using criteria such as Information Gain or Gini Index.
3. **Split the Dataset:** Divide the dataset into subsets based on feature values.
 - If the Age > 30, split into two groups (greater than 30 / less than or equal to 30).
 - If the Petal_length < 2.45, split into two groups (greater than 2.45 / less than or equal to 2.45).
4. **Create Decision Nodes and Branches:** Each split creates a decision node, connected by branches that represent decision rules.
5. **Repeat Recursively:** Continue selecting features and splitting until reaching stopping criteria such as:
 - Maximum tree depth
 - Too few samples in a node
 - No feature provides further separation
6. **Leaf Nodes:** When splitting stops, the final node becomes a leaf node that outputs the result.

Best Splitting Criteria

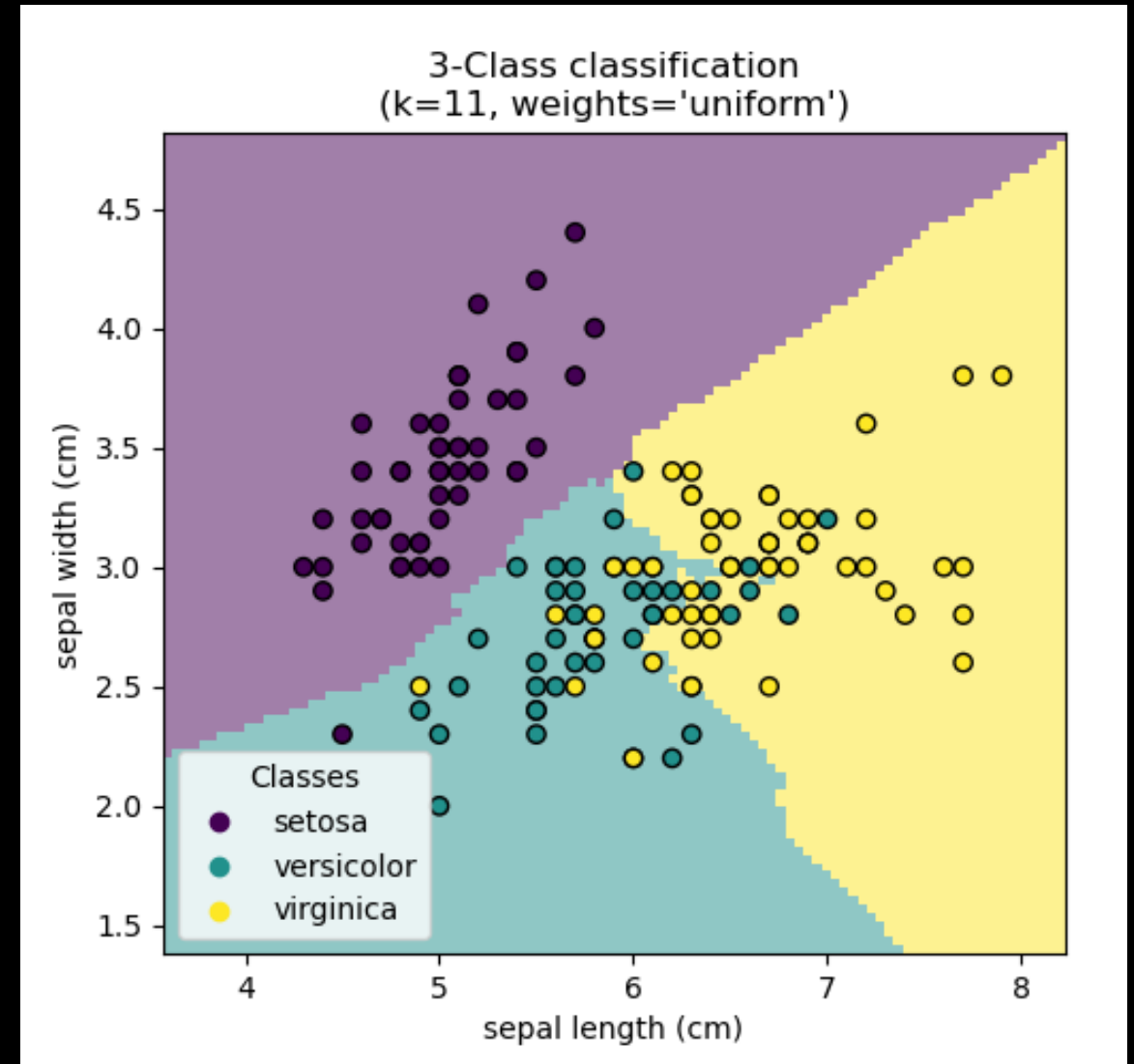
- Goal: At each step, the tree must decide which feature (variable) will split the data most effectively into distinct classes.
- Criteria: Common mathematical measures are used to evaluate how “good” a split is:
 - Information Gain (Entropy-based): Measures reduction in uncertainty after the split.
 - Gini Index: Measures impurity; lower Gini means purer subsets.
 - Chi-Square Test: Sometimes used for categorical data.
- Process
 - Calculate the impurity (entropy or Gini) of the current dataset.
 - For each candidate feature, simulate splitting the dataset by its values.
 - Compute the impurity of each subset.
 - Calculate the weighted average impurity after the split.
 - The feature that gives the largest reduction in impurity (highest Information Gain or lowest Gini) is chosen.

Nearest Neighbors

A type of instance-based learning or non-generalizing learning: it does not attempt to construct a general internal model, but simply stores instances of the training data.

Components:

- **Training Data:** Labeled data containing features and target classes.
- **Query Point:** A new data point that needs to be classified.
- **K-Value:** The number of nearest neighbors considered for prediction.
- **Distance Metric:** A method for measuring the distance between data points, such as Euclidean distance.
- **Voting:** The classes of the selected neighbors are compared, and the majority class is used for classification.
- **Feature Scaling:** Features should generally be scaled because KNN relies on distance calculations.



<https://scikit-learn.org/stable/modules/neighbors.html>

Nearest Neighbors (Cont.)

Training Data

Student	Math	Science	Class	Distance	Nearest
A	2	3	B		
B	3	4	B		
C	4	5	A		
D	6	7	A		
E	7	8	A		

New data $X = (5, 5)$

And $K = 3$



Select the 3 closest points:

C → Class A

B → Class B

D → Class A

→ A

1. Calculate the Euclidean Distance

Formula:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

For the new data point $X = (5, 5)$:

A = (2,3)

$$d = \sqrt{(5 - 2)^2 + (5 - 3)^2} = \sqrt{9 + 4} = \sqrt{13} \approx 3.61$$

B = (3,4)

$$d = \sqrt{(5 - 3)^2 + (5 - 4)^2} = \sqrt{4 + 1} = \sqrt{5} \approx 2.24$$

C = (4,5)

$$d = \sqrt{(5 - 4)^2 + (5 - 5)^2} = 1$$

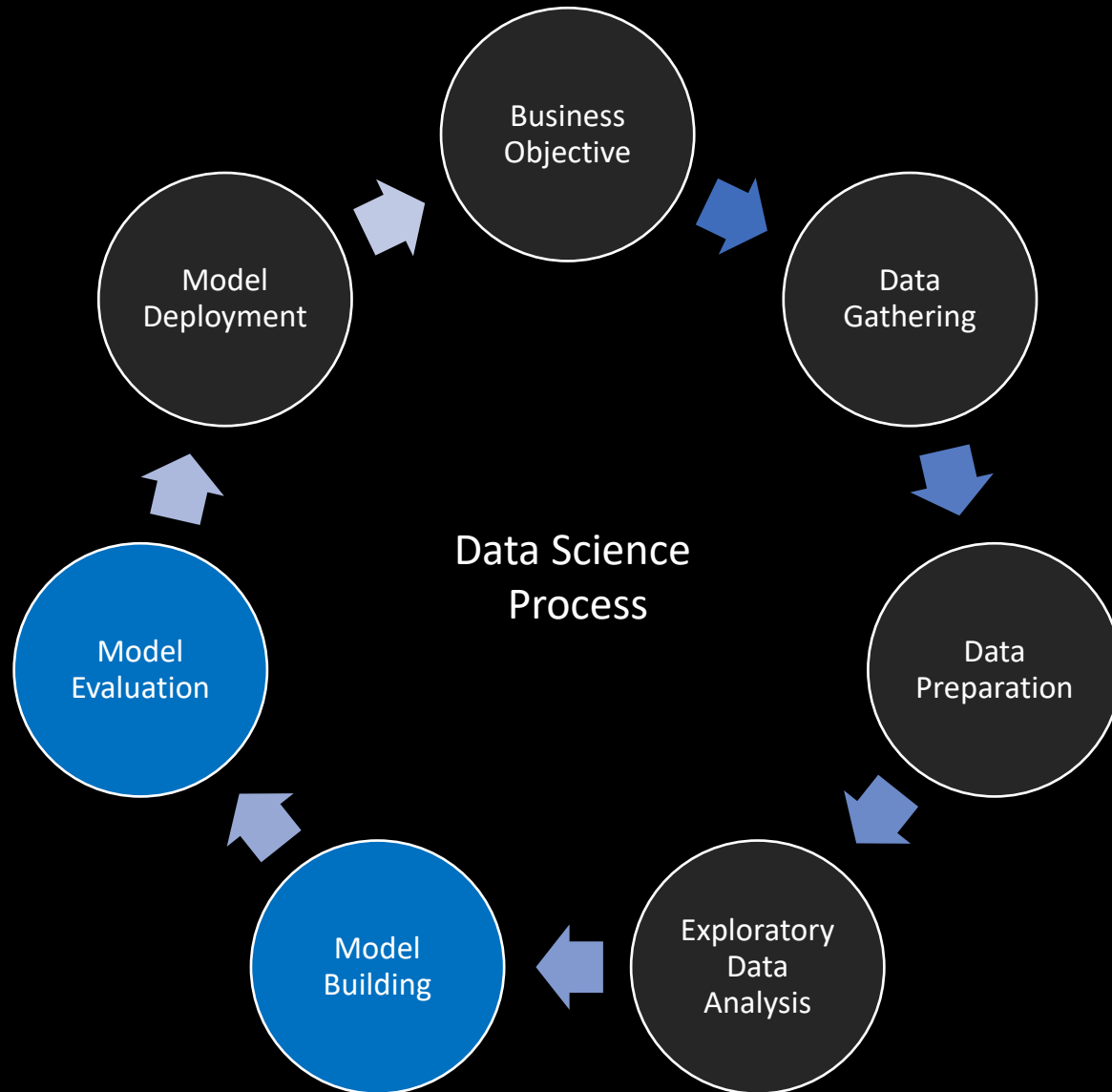
D = (6,7)

$$d = \sqrt{(5 - 6)^2 + (5 - 7)^2} = \sqrt{5} \approx 2.24$$

E = (7,8)

$$d = \sqrt{(5 - 7)^2 + (5 - 8)^2} = \sqrt{13} \approx 3.61$$

Data Science Process



Business Objective

- Should the problem be solved with machine learning?

Data Gathering

- Can we gather the example data for the prediction model?

Data Preparation

- Are the gathered data clean enough to create the prediction model?

Exploratory Data Analysis

- What features are needed to create the prediction model?

Model Building

- What type of the base model (algorithm) we will use?

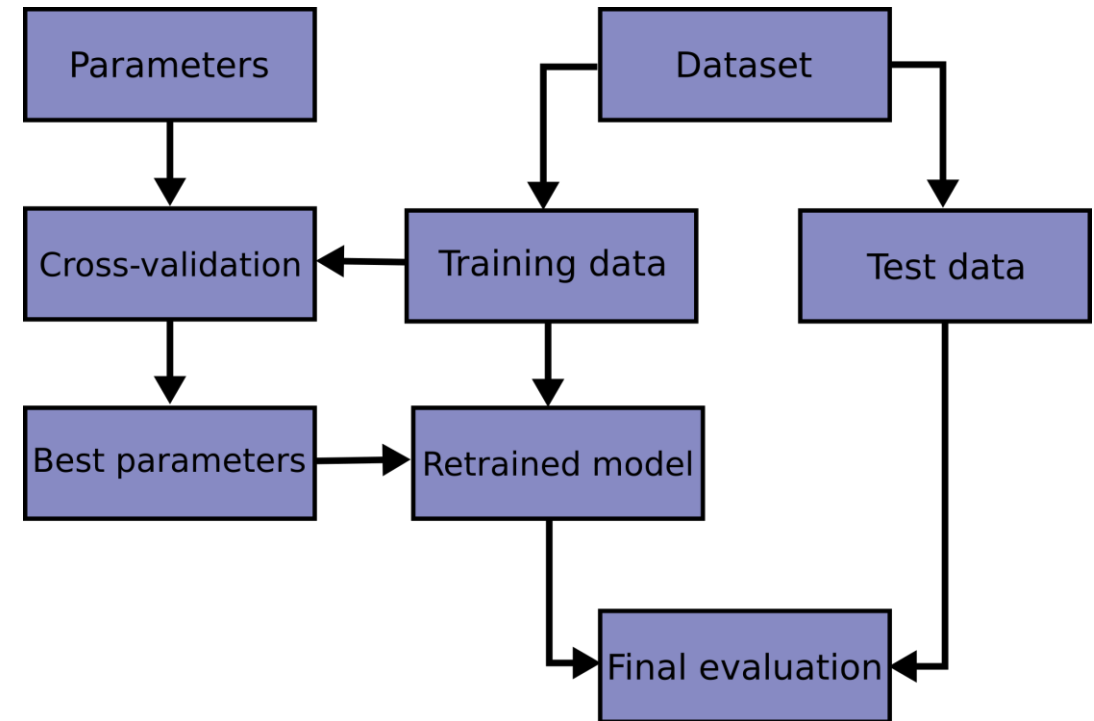
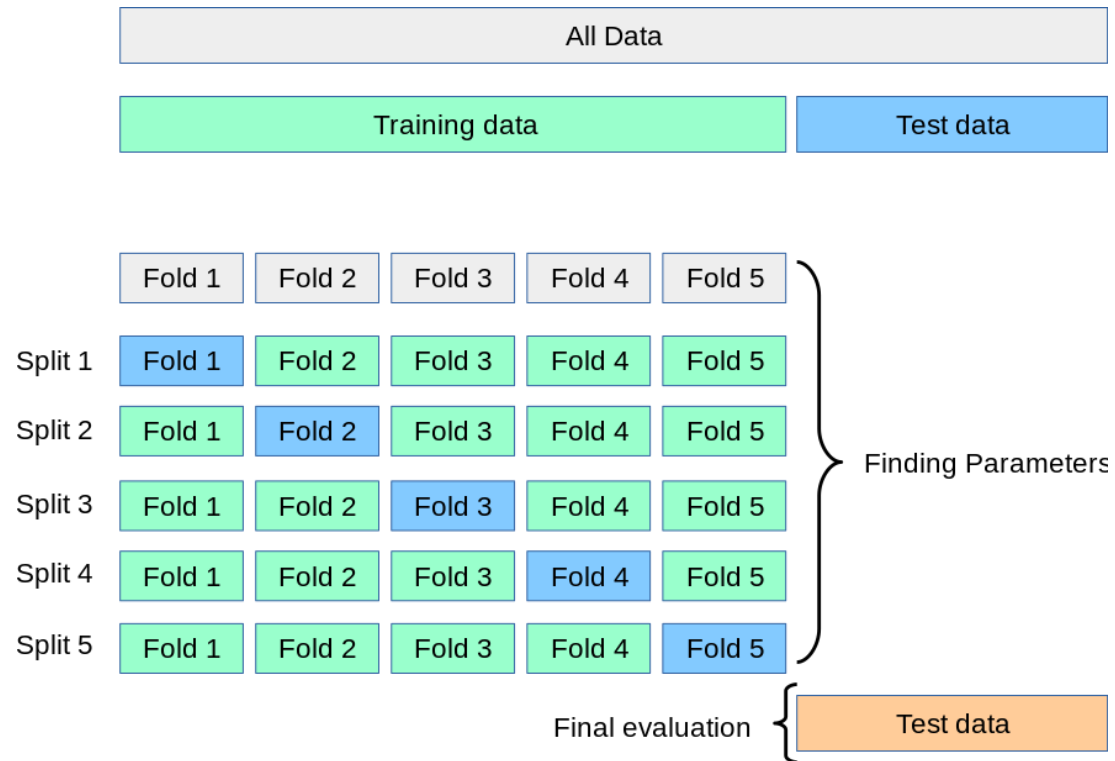
Model Evaluation

- Does the model provide the acceptable accuracy?

Model Deployment

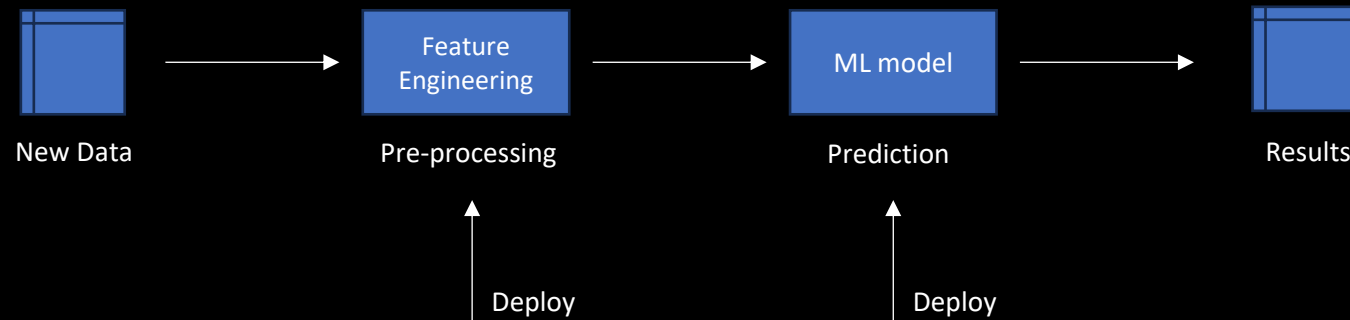
- How to take the trained prediction model to production?

How to develop classification applications?

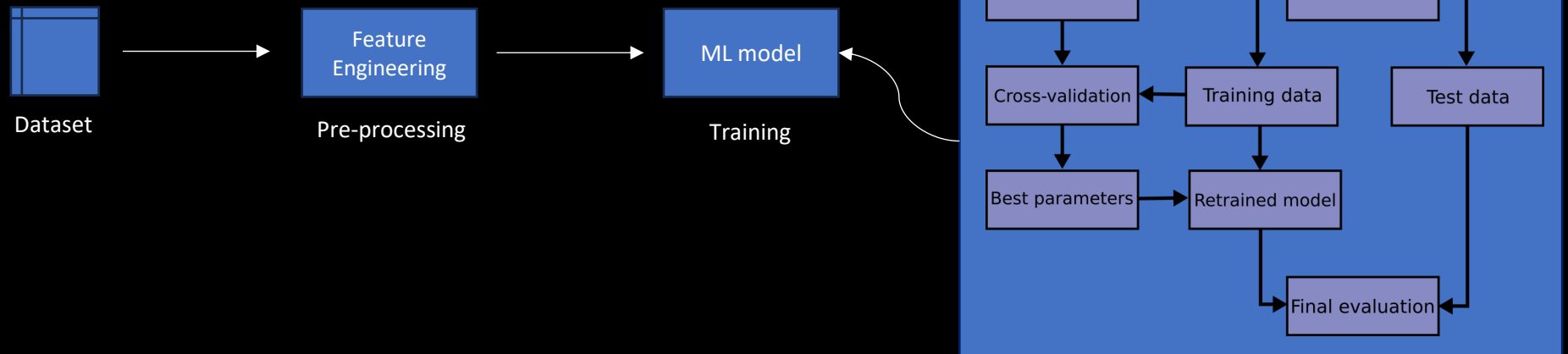


How to develop classification applications?

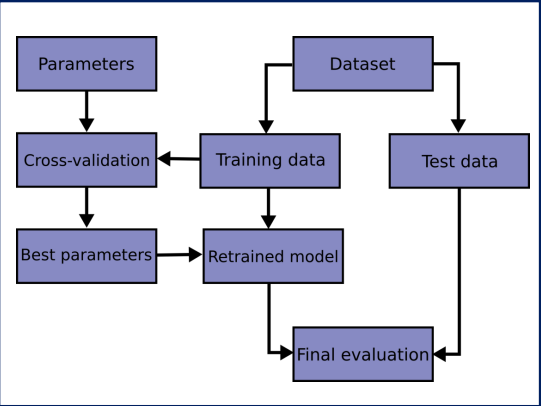
Using Phase



Training Phase



Building Classification Model (Cont.)



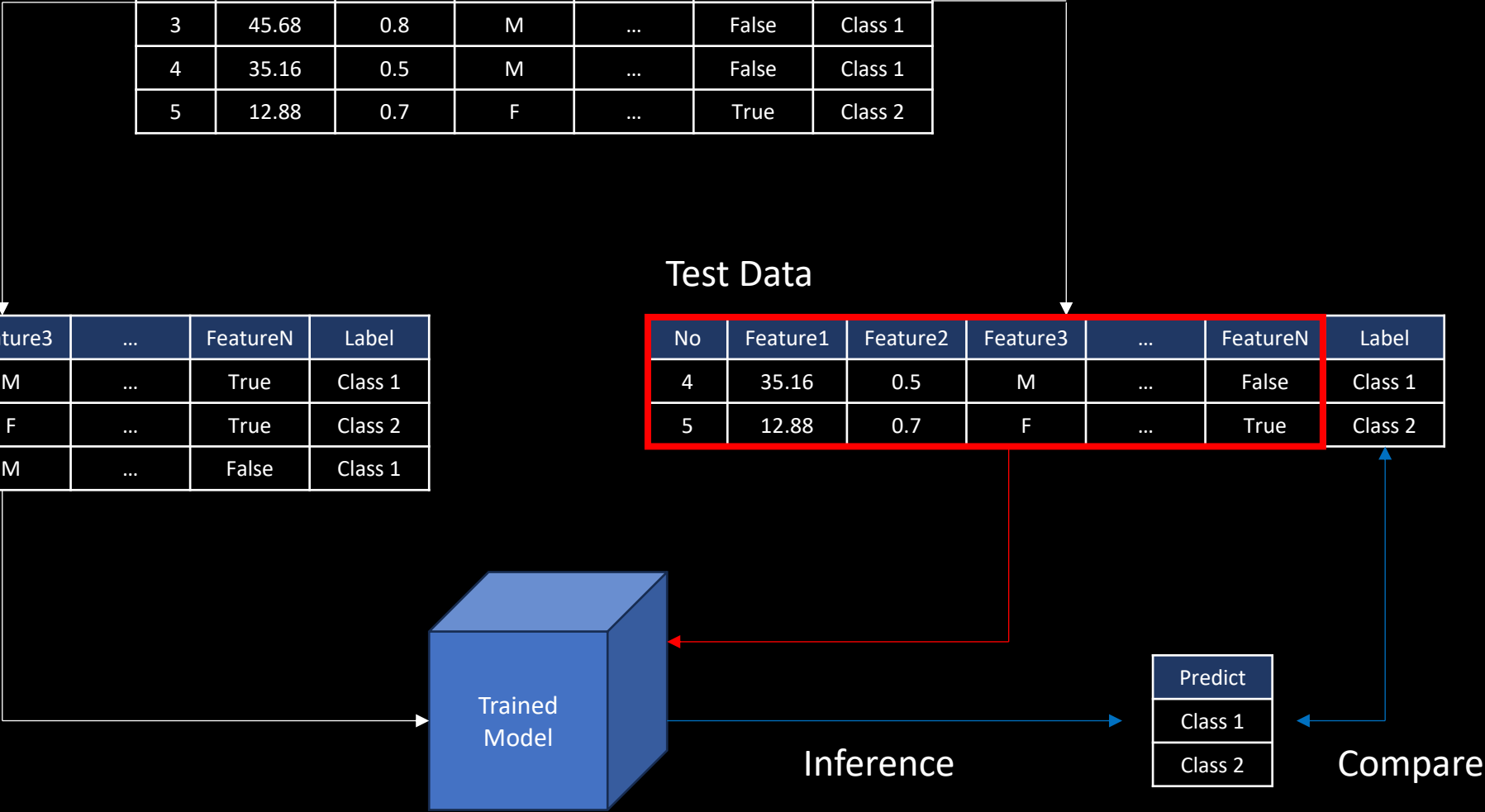
No	Feature1	Feature2	Feature3	...	FeatureN	Label
1	12.87	0.5	M	...	True	Class 1
2	15.99	0.3	F	...	True	Class 2
3	45.68	0.8	M	...	False	Class 1
4	35.16	0.5	M	...	False	Class 1
5	12.88	0.7	F	...	True	Class 2

Training Data

No	Feature1	Feature2	Feature3	...	FeatureN	Label
1	12.87	0.5	M	...	True	Class 1
2	15.99	0.3	F	...	True	Class 2
3	45.68	0.8	M	...	False	Class 1

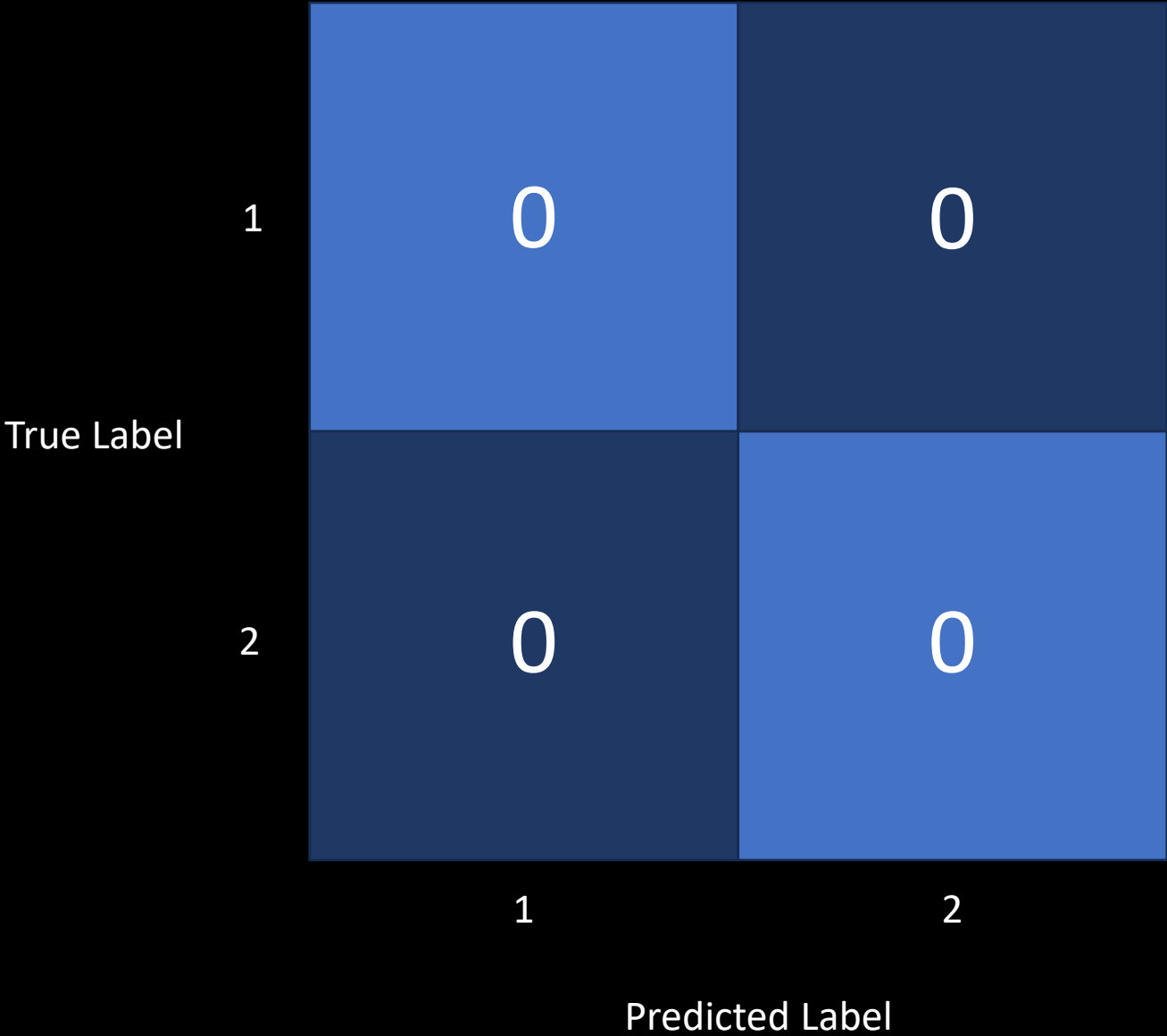
Test Data

No	Feature1	Feature2	Feature3	...	FeatureN	Label
4	35.16	0.5	M	...	False	Class 1
5	12.88	0.7	F	...	True	Class 2

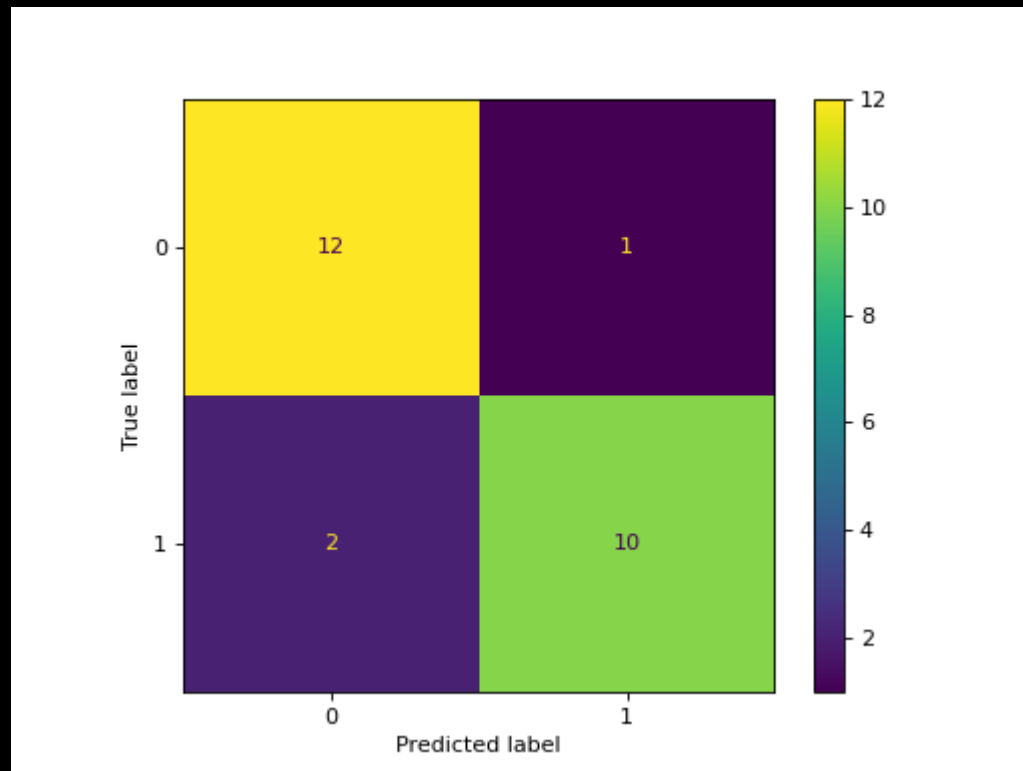


Building Classification Model (Cont.)

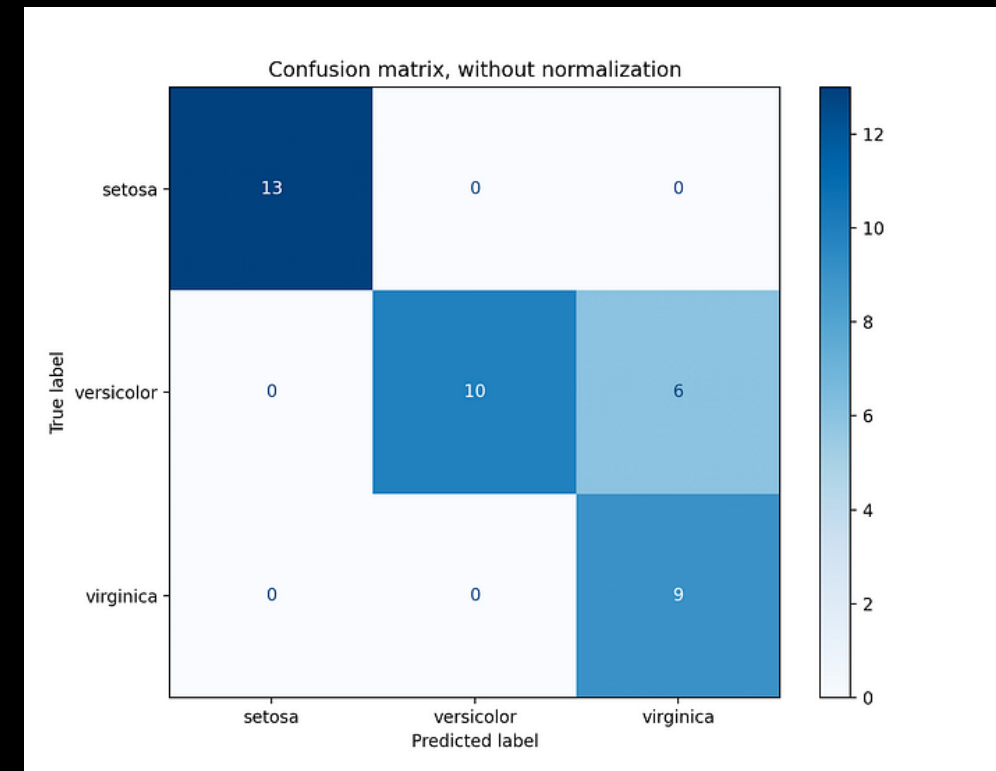
No	Predict	Label
1	Class 1	Class 1
2	Class 2	Class 2
3	Class 1	Class 2
4	Class 2	Class 2
5	Class 1	Class 1
6	Class 2	Class 2
7	Class 2	Class 2
8	Class 1	Class 1
9	Class 1	Class 1
10	Class 1	Class 2
11	Class 2	Class 2
12	Class 2	Class 2
13	Class 2	Class 1
14	Class 1	Class 1
15	Class 1	Class 1
16	Class 2	Class 2
17	Class 2	Class 2
18	Class 1	Class 2
19	Class 1	Class 1
20	Class 2	Class 1



How to evaluate the performance classifiers?

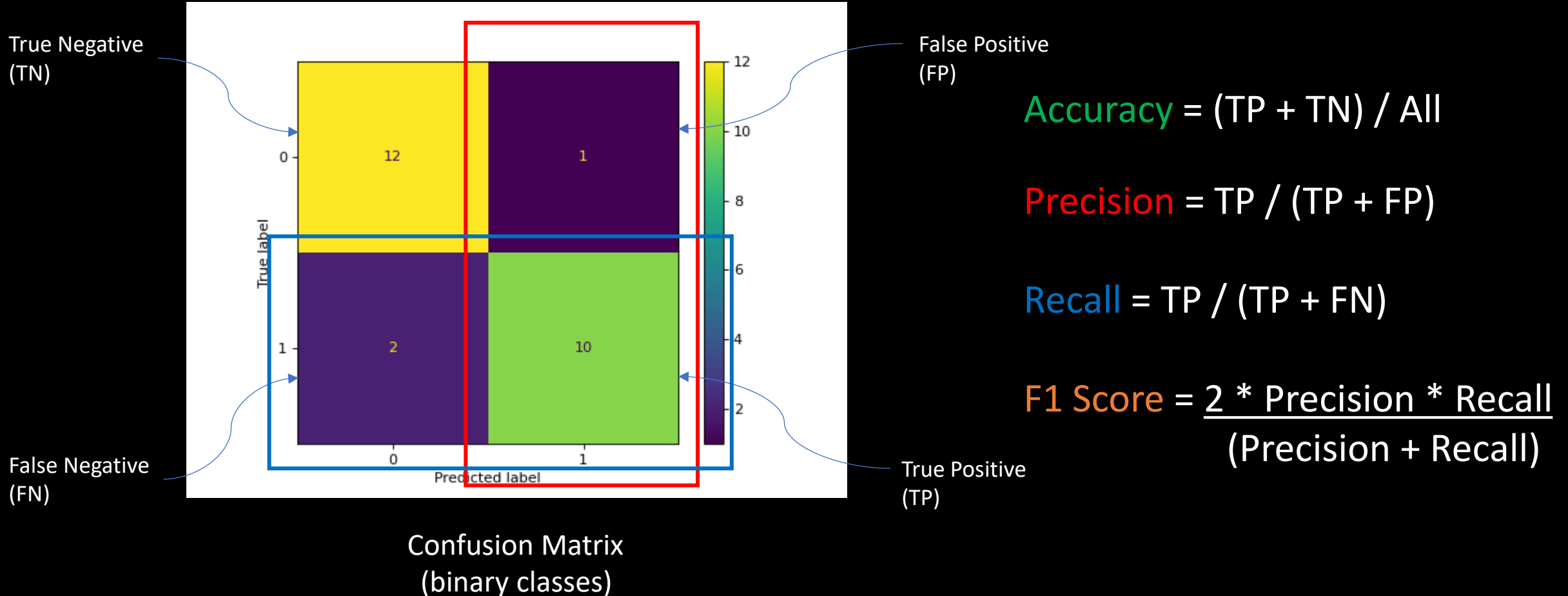


Confusion Matrix
(binary classes)

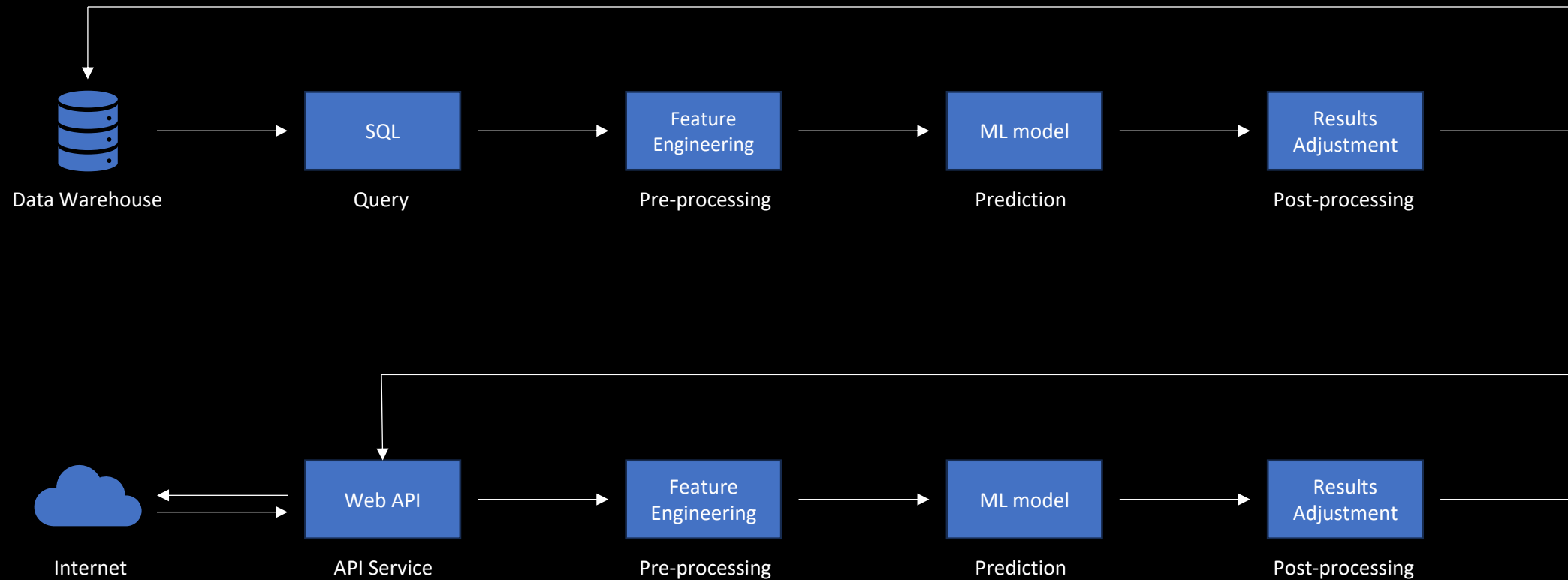


Confusion Matrix
(Multi classes)

How to evaluate the performance classifiers?



- How to develop classification applications?

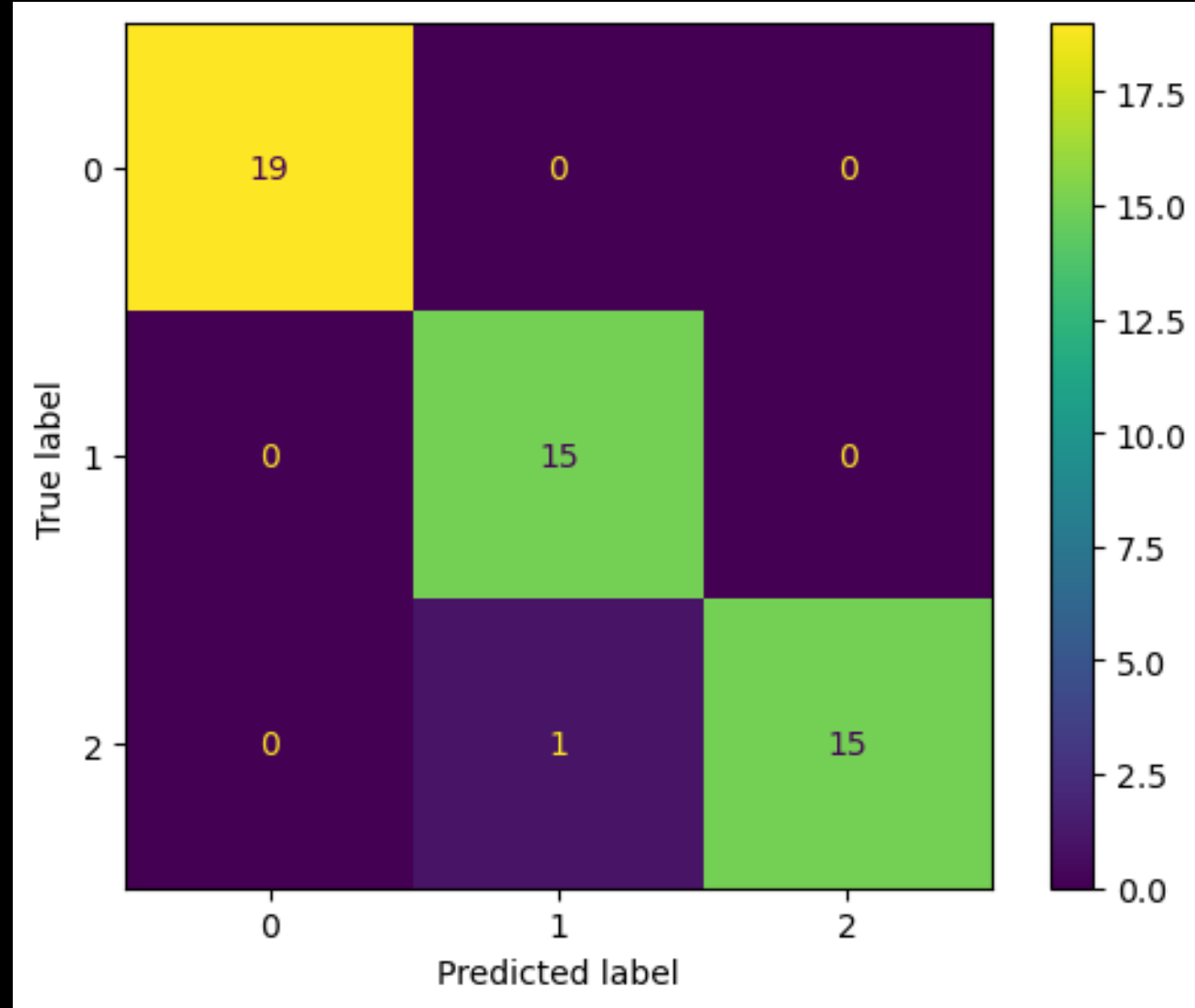


Example 1: Iris Dataset

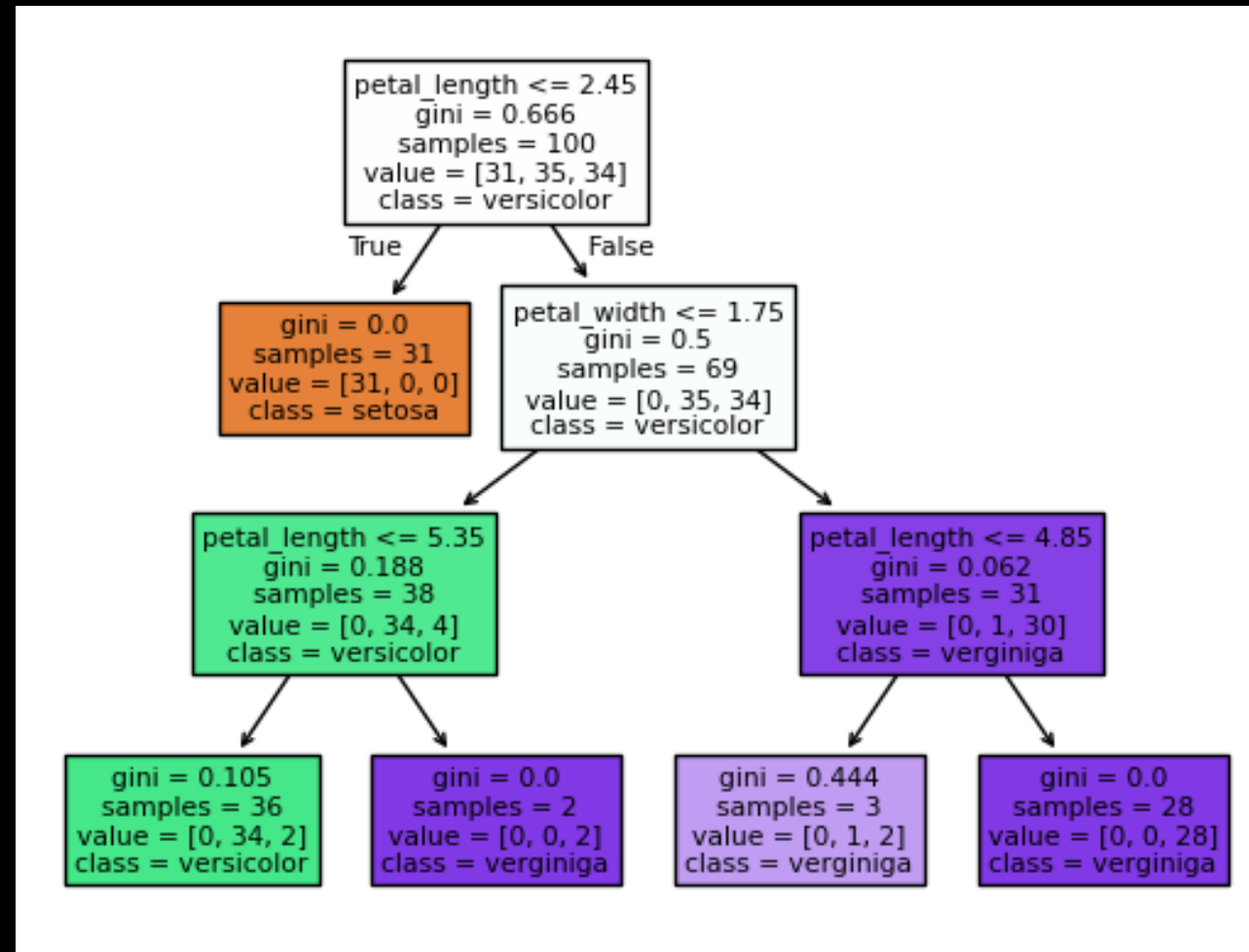
- Given Iris Dataset, create a classification model



Example 1: Iris Dataset (Cont.)

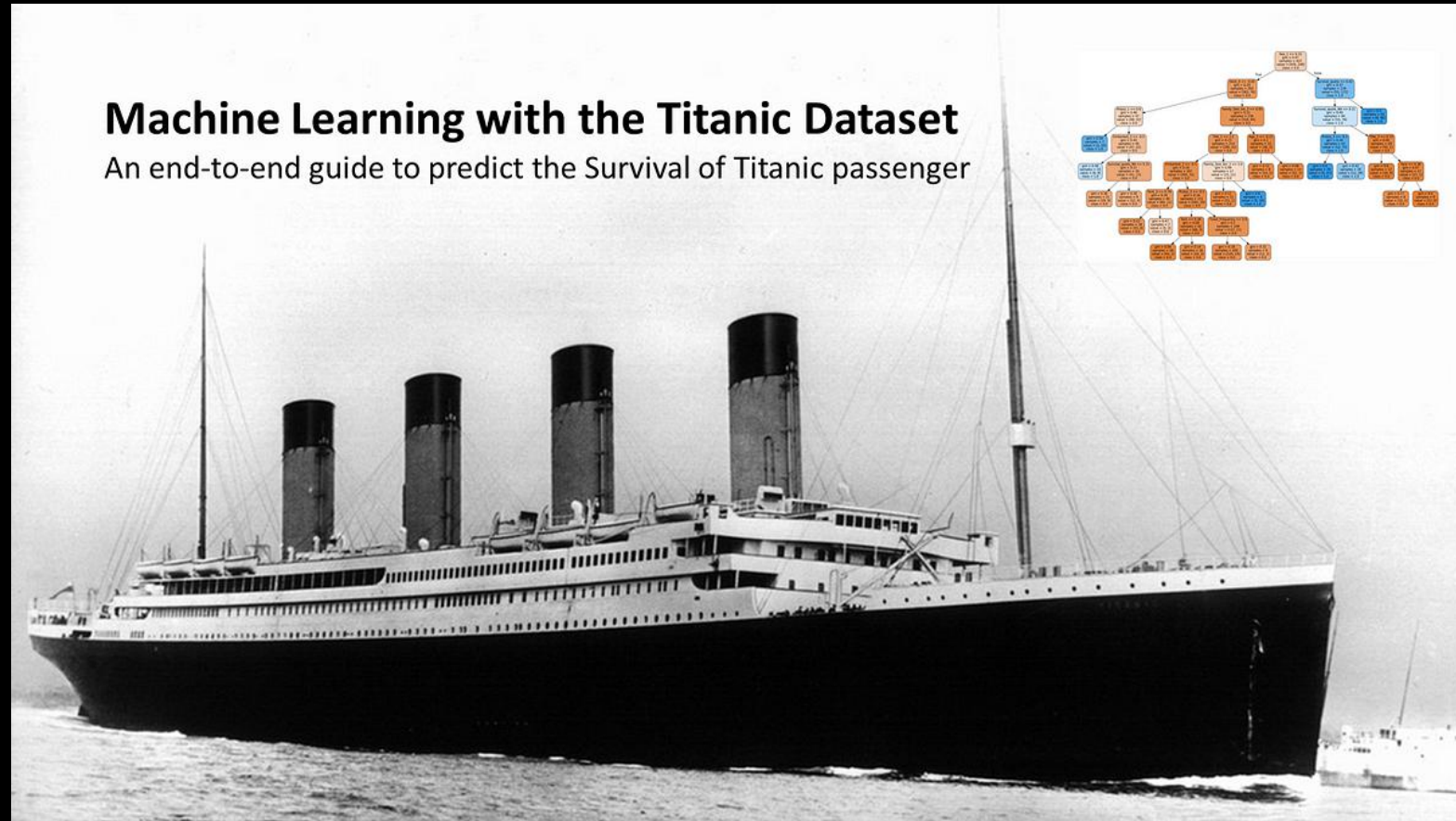


Example 1: Iris Dataset (Cont.)



Example 2 : Titanic Dataset

Given Titanic Dataset, create a classification model

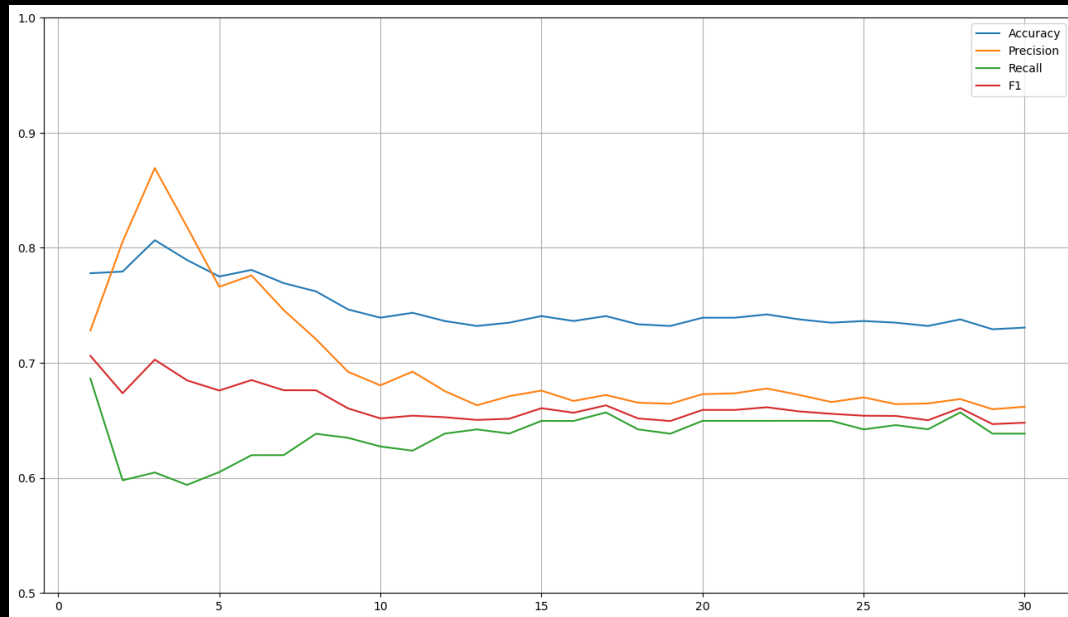


Features

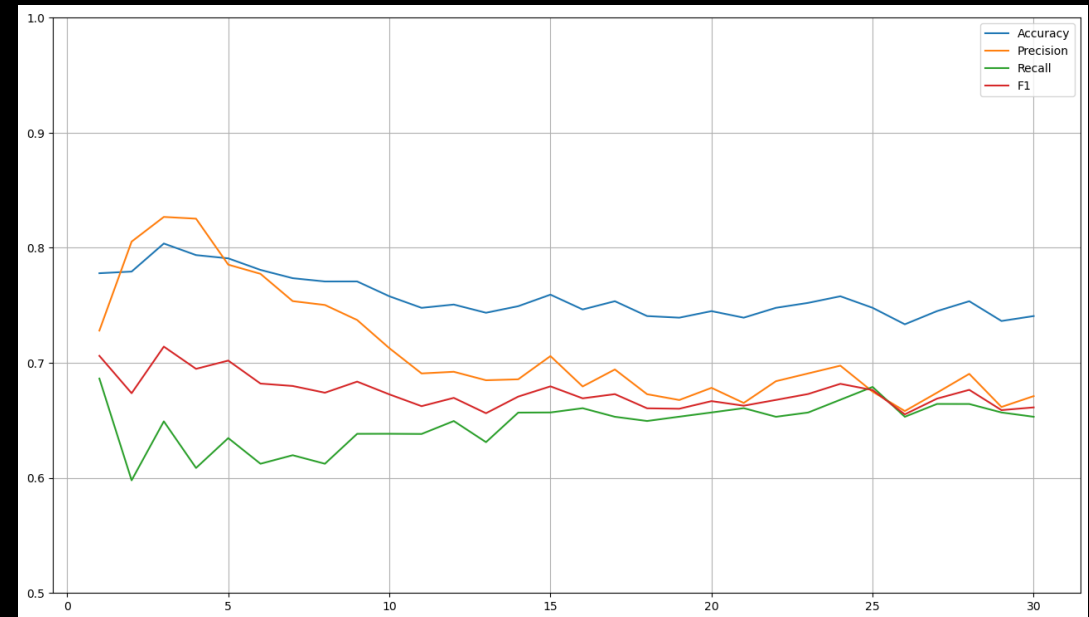
1. survival - Survival (0 = No; 1 = Yes)
2. class - Passenger Class (1 = 1st; 2 = 2nd; 3 = 3rd)
3. name - Name
4. sex - Sex
5. age - Age
6. sibsp - Number of Siblings/Spouses Aboard
7. parch - Number of Parents/Children Aboard
8. ticket - Ticket Number
9. fare - Passenger Fare
10. cabin - Cabin
11. embarked - Port of Embarkation (C = Cherbourg; Q = Queenstown; S = Southampton)
12. boat - Lifeboat (if survived)
13. body - Body number (if did not survive and body was recovered)

Example 2 : Titanic Dataset (Cont.)

GINI



Entropy



Given Breast cancer dataset , create a classification model

Home Work !?

Can you answer the following questions ?

☐ What is classification problem?

- To classify new data into defined classes (Yes/No, Survived/Died, or Type of flowers).

☐ Why can machine learning solve classification problems?

- It can recognize relationships between features and label in the training data.

☐ What techniques and tools used to solve the problem?

- Decision Trees, Nearest Neighbors, etc.

☐ How to create classification models for the problems?

- Using Scikit-learn library is a popular way.

☐ Is my classification model good enough?

- Measured by Accuracy, Precision and Recall values